About this document

This is the assignment for the UCL CASA module Data Science for Spatial Systems (CASA0006). This is a self-contained Jupyter notebook with structured introduction, methodology, analysis and bibliography.

Data is available here

Two-Wheeler Collision Severity in London: A Spatio-Temporal Analysis

Using Machine Learning to Identify Risk Factors and Safety Intervention Priorities

This study examines how temporal patterns, road infrastructure characteristics, and fine-grained land use environments jointly influence the severity of two-wheeler collisions in London, using DfT 2025 provisional road safety data enriched with OpenStreetMap land use information.

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1.0 Introduction

Road traffic collisions remain a significant public health challenge in the UK. In 2024, 128,272 casualty accidents were reported, of which 29,467 resulted in death or serious injury (DfT, 2025). Vulnerable road users bear a disproportionate share of this risk: two-wheeled riders accounted for 26% of road traffic fatalities in 2024, with motorcyclist fatalities increasing by 8% (DfT, 2025). The rapid growth of gig economy food delivery platforms has significantly increased two-wheeler presence on urban streets, with gig riders nearly twice as likely to report collision injuries compared to employed riders due to per-delivery pay structures incentivising speed (Christie & Ward, 2023). Despite growing investment in cycling infrastructure, cyclists and motorcyclists remain among the highest-risk road users, with the smallest reduction in fatality rates among all road user groups in recent years (European Commission, 2025).

Research consistently identifies temporal and spatial factors as key predictors of two-wheeler collision severity. Night-time crashes are associated with greater injury severity (Jingjing et al., 2026), while road type, junction characteristics, and

land use context all influence collision outcomes (Xiao et al., 2025; Antonella et al., 2023). However, existing studies rarely integrate fine-grained land use data with temporal and infrastructure factors in a unified predictive framework. A study of motorcycle crashes in Thailand demonstrated that random forest outperformed seven other algorithms in predicting fatal outcomes (Mohamad et al., 2025). Machine learning methods, particularly random forest classifiers combined with SHAP interpretability tools, have demonstrated strong performance in collision severity modelling (Antonella et al., 2023; Antonella et al., 2025).

This study uses DfT 2025 provisional road safety statistics, spatially joined with OpenStreetMap land use data, to examine how temporal patterns, road infrastructure, and land use environments jointly influence two-wheeler collision severity in London, and to identify which factors offer the greatest potential for safety interventions.

2.0 Research Questions

RQ1: What factors are associated with the severity of two-wheeler collisions in London?

RQ2: Which factors contribute most to predicting serious or fatal collision outcomes, and what do they imply for road safety interventions?

```
# Import Libraries

!pip install imbalanced-learn shap osmium --quiet

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import seaborn as sns
import geopandas as gpd
import zipfile, io, requests
import osmium
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, roc_auc_score, RocCurveD
from sklearn.preprocessing import LabelEncoder
from statsmodels.stats.outliers_influence import variance_inflation_factor
from imblearn.over_sampling import SMOTE
import shap
import warnings
warnings.filterwarnings('ignore')
```

```
[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
```

3.0 Data

This study used three datasets from the UK Department for Transport 2025 road safety statistics: a collision table, a vehicle table, and an injury table.

```
# 3.1 Load and Merge Data

collision_url = "https://raw.githubusercontent.com/siyi-w/DSSS/main/as/data
vehicle_url   = "https://raw.githubusercontent.com/siyi-w/DSSS/main/as/data
casualty_url  = "https://raw.githubusercontent.com/siyi-w/DSSS/main/as/data

df_collision = pd.read_csv(collision_url)
df_vehicle   = pd.read_csv(vehicle_url)
df_casualty  = pd.read_csv(casualty_url)

print(f"Collision records : {len(df_collision):,}")
print(f"Vehicle records   : {len(df_vehicle):,}")
print(f"Casualty records  : {len(df_casualty):,}")
```

```
Collision records : 48,472
Vehicle records   : 87,805
Casualty records  : 60,991
```

This study focuses on two-wheeled vehicle collisions in London.

```
# 3.2 Data Cleaning, Filtering and Feature Engineering

# 3.2.1 Filter London (Metropolitan Police=1, City of London=2)
# First, accidents were filtered to the Metropolitan Police and the City of
# Resulting in 10,062 collisions in London.

df_london = df_collision[df_collision['police_force'].isin([1, 2])].copy()
print(f"London collisions: {len(df_london):,}")
```

```
London collisions: 10,062
```

```
#  3.2.2 Filter two-wheelers and merge vehicle_type

#Then, the vehicle table was used to identify collisions involving two-whee
# defined as bicycles, motorcycles with an engine displacement of no more t

two_wheeler_types = [1, 2, 3, 4, 5, 23]
df_tw_vehicle = df_vehicle[df_vehicle['vehicle_type'].isin(two_wheeler_type
tw_ids = df_tw_vehicle['collision_index'].unique()
df_tw = df_london[df_london['collision_index'].isin(tw_ids)].copy()
print(f"London two-wheeler collisions: {len(df_tw):,}")

df_tw = df_tw.merge(
    df_tw_vehicle[['collision_index', 'vehicle_type']].drop_duplicates('col
    on='collision_index',
    how='left'
)

print("Vehicle type distribution:")
print(df_tw['vehicle_type'].value_counts())
```

```
London two-wheeler collisions: 4,966
Vehicle type distribution:
vehicle_type
1     2693
3     1601
5      359
4      184
2       72
23      57
Name: count, dtype: int64
```

The outcome variable is a binary collision severity , with serious or fatal accidents coded as 1 and minor injuries coded as 0. The final class distribution is approximately 80% minor injuries and 20% serious or fatal accidents, reflecting the class imbalance problem addressed by applying SMOTE oversampling only to the training set.

```
#  3.2.3 Binary severity target

df_tw['severity_binary'] = (df_tw['collision_severity'] <= 2).astype(int)
print(f"\nSeverity distribution:")
print(df_tw['severity_binary'].value_counts(normalize=True).round(3))

#  3.2.4 Time features

df_tw['date'] = pd.to_datetime(df_tw['date'], dayfirst=True)
df_tw['hour'] = df_tw['time'].str[:2].astype(float)
df_tw['is_weekend'] = (df_tw['day_of_week'] >= 5).astype(int)
```

```
Severity distribution:
severity_binary
0    0.796
1    0.204
Name: proportion, dtype: float64
```

To capture land use context, London's OpenStreetMap land use polygons are spatially connected to the collision points.

```
#  3.3 OSM Landuse Spatial Join (50m buffer)

#Since collisions on roads often occur outside of land use polygons, the ma
# Therefore, a 50-meter buffer is applied to unmatched points, improving th
# Land use categories are simplified to five: residential, commercial, gree

zip_url = "https://raw.githubusercontent.com/siyi-w/DSSS/main/as/data/landu
r = requests.get(zip_url)
z = zipfile.ZipFile(io.BytesIO(r.content))
z.extractall("/tmp/landuse/")

landuse = gpd.read_file("/tmp/landuse/gis_osm_landuse_a_free_1.shp",
                        columns=['fclass', 'geometry'])

landuse_map = {
    'retail': 'commercial', 'commercial': 'commercial',
    'industrial': 'industrial', 'residential': 'residential',
    'park': 'green', 'grass': 'green', 'forest': 'green',
    'meadow': 'green', 'scrub': 'green',
```

```
    'recreation_ground': 'green', 'farmland': 'green',
    'farmyard': 'green', 'heath': 'green',
    'allotments': 'green', 'cemetery': 'green',
}
landuse['landuse_class'] = landuse['fclass'].map(landuse_map).fillna('other

landuse = landuse[landuse['landuse_class'] != 'other'].copy()
print(f"Landuse polygons (filtered): {len(landuse):,}")

gdf_tw = gpd.GeoDataFrame(
    df_tw.copy(),
    geometry=gpd.points_from_xy(df_tw['longitude'], df_tw['latitude']),
    crs='EPSG:4326'
).to_crs(epsg=27700)
landuse = landuse.to_crs(epsg=27700)
joined = gpd.sjoin(gdf_tw, landuse[['geometry', 'landuse_class']],
                   how='left', predicate='within')
joined = joined[~joined.index.duplicated(keep='first')]
unmatched = joined[joined['landuse_class'].isna()].index
print(f"Direct match: {joined['landuse_class'].notna().sum():,}, Unmatched:

if len(unmatched) > 0:
    gdf_un = gdf_tw.loc[unmatched].copy()
    gdf_un['geometry'] = gdf_un.geometry.buffer(50)
    joined_buf = gpd.sjoin(gdf_un, landuse[['geometry', 'landuse_class']],
                           how='left', predicate='intersects')
    joined_buf = joined_buf[~joined_buf.index.duplicated(keep='first')]
    joined.loc[unmatched, 'landuse_class'] = joined_buf['landuse_class']

joined['landuse_class'] = joined['landuse_class'].fillna('other')
df_tw['landuse_class'] = joined['landuse_class'].values
print(f"\nLanduse distribution:")
print(df_tw['landuse_class'].value_counts())
```

```
Landuse polygons (filtered): 79,672
Direct match: 1,035, Unmatched: 3,931

Landuse distribution:
landuse_class
residential    2981
commercial     1046
green           697
industrial      124
other           118
Name: count, dtype: int64
```

To capture the intensity of gig economy delivery and use it as an indicator of two-wheeler exposure, we calculate the food density at each collision point. The Greater London OSM file is available at: https://download.geofabrik.de/europe/great-britain/england/greater-london.html

```
#  3.4 Food Establishment Density (500m buffer)

# Locations of restaurants, cafes, fast food outlets, and food courts (amen
# Calculated the number of food establishments within a 500-meter buffer zo

food_density_url = "https://raw.githubusercontent.com/siyi-w/DSSS/main/as/d
```

```
pbf_path = '/Users/wusiyi/Documents/CASA/dsss_data/greater-london-260424.os

if 'food_density' in df_tw.columns:
    df_tw = df_tw.drop(columns=['food_density'])

food_loaded = False

try:
    food_density_pre = pd.read_csv(food_density_url)
    df_tw = df_tw.merge(food_density_pre, on='collision_index', how='left')
    df_tw['food_density'] = df_tw['food_density'].fillna(0)
    print(f"Food density loaded from GitHub, mean: {df_tw['food_density'].m
    food_loaded = True

except Exception as e:
    print(f"GitHub CSV not found ({e}), will compute from local pbf...")

if not food_loaded:
    class FoodHandler(osmium.SimpleHandler):
        def __init__(self):
            super().__init__()
            self.foods = []

        def node(self, n):
            amenity = n.tags.get('amenity', '')
            if amenity in ['restaurant', 'cafe', 'fast_food', 'food_court']
                self.foods.append({
                    'lat': n.location.lat,
                    'lon': n.location.lon
                })

    handler = FoodHandler()
    handler.apply_file(pbf_path)
    df_food = pd.DataFrame(handler.foods)

    gdf_food = gpd.GeoDataFrame(
        df_food,
        geometry=gpd.points_from_xy(df_food['lon'], df_food['lat']),
        crs='EPSG:4326'
    ).to_crs(epsg=27700)

    gdf_accidents = gpd.GeoDataFrame(
        df_tw.copy(),
        geometry=gpd.points_from_xy(df_tw['longitude'], df_tw['latitude']),
        crs='EPSG:4326'
    ).to_crs(epsg=27700)

    gdf_accidents_buf = gdf_accidents.copy()
    gdf_accidents_buf['geometry'] = gdf_accidents_buf.geometry.buffer(500)

    joined_food = gpd.sjoin(
        gdf_accidents_buf[['collision_index', 'geometry']],
        gdf_food[['geometry']],
        how='left',
        predicate='contains'
    )

    food_density = joined_food.groupby('collision_index').size().reset_inde
    food_density.columns = ['collision_index', 'food_density']
```

```
    df_tw = df_tw.merge(food_density, on='collision_index', how='left')
    df_tw['food_density'] = df_tw['food_density'].fillna(0)
    df_tw[['collision_index', 'food_density']].to_csv(
        'food_density_precomputed.csv', index=False)
    print(f"Food density computed and saved, mean: {df_tw['food_density'].m
```

```
Food density loaded from GitHub, mean: 27.4
```

```
#  3.5 One-Hot Encoding for Landuse

landuse_dummies = pd.get_dummies(df_tw['landuse_class'], prefix='lu').astyp
df_tw = pd.concat([df_tw.reset_index(drop=True),
                   landuse_dummies.reset_index(drop=True)], axis=1)
landuse_cols = [c for c in df_tw.columns if c.startswith('lu_')]
landuse_cols = [c for c in landuse_cols if c != 'lu_commercial']  # commerc
print(f"Landuse dummy columns: {landuse_cols}")
```

```
Landuse dummy columns: ['lu_green', 'lu_industrial', 'lu_other', 'lu_resid
ential']
```

Vehicle type was retained as a predictor to distinguish between bicycles, motorcycles, and electric two-wheelers, as their severity differed significantly. Multicollinearity was assessed using the variance inflation factor.

```
#  3.6 VIF Check and Feature Selection

features_vif = ['hour', 'is_weekend',
                'road_type', 'first_road_class', 'speed_limit', 'junction_d
                'light_conditions', 'weather_conditions',
                'road_surface_conditions', 'urban_or_rural_area',
                'number_of_vehicles', 'number_of_casualties',
                'vehicle_type', 'food_density'] + landuse_cols

df_vif = df_tw[features_vif].dropna().copy()
vif_data = pd.DataFrame({
    'Feature': features_vif,
    'VIF': [variance_inflation_factor(df_vif.values.astype(float), i)
            for i in range(len(features_vif))]
})
print("\nVIF Results:")
print(vif_data.sort_values('VIF', ascending=False).to_string())

#Since urban_or_rural_area had a VIF of 35.3, it was excluded from the fina
```

```
VIF Results:
                    Feature        VIF
9       urban_or_rural_area  35.307557
10       number_of_vehicles  20.875775
11     number_of_casualties  20.180986
4               speed_limit  15.103528
3          first_road_class  11.423762
0                      hour   8.929185
2                 road_type   8.076101
17           lu_residential   4.342748
8   road_surface_conditions   3.075559
7        weather_conditions   2.910444
6          light_conditions   2.722518
1                is_weekend   1.883935
12             vehicle_type   1.857247
14                 lu_green   1.843489
13             food_density   1.663174
5           junction_detail   1.643165
16                 lu_other   1.186985
15            lu_industrial   1.161277
```

```python
#  3.7 Feature Selection
#   Remove urban_or_rural_area due to extreme multicollinearity (VIF=35.3)

features = ['hour', 'is_weekend',
            'road_type', 'first_road_class', 'speed_limit', 'junction_detai
            'light_conditions', 'weather_conditions',
            'road_surface_conditions',
            'number_of_vehicles', 'number_of_casualties',
            'vehicle_type', 'food_density'] + landuse_cols

df_model = df_tw[features + ['severity_binary']].dropna()
print(f"\nFinal modelling dataset: {len(df_model):,} rows")
print(f"Serious/Fatal rate: {df_model['severity_binary'].mean():.1%}")
print(f"Number of features: {len(features)}")
```

```
Final modelling dataset: 4,966 rows
Serious/Fatal rate: 20.4%
Number of features: 17
```

4.0 Methodolgy

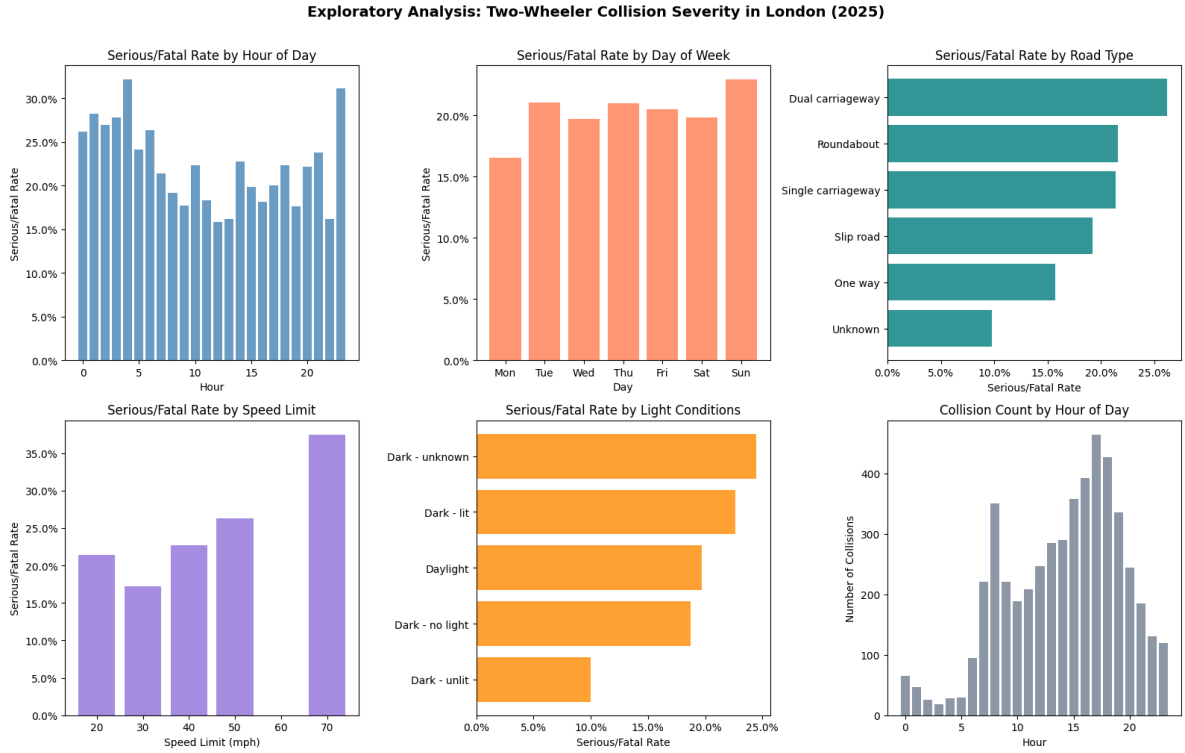
This study employs four analytical methods to address the research questions sequentially.

First, Exploratory Data Analysis including a correlation heatmap was conducted to identify patterns in collision severity across temporal, road infrastructure, and land use dimensions, guiding feature selection for subsequent modelling. OSM land use polygons were initially joined directly to collision points, achieving only a 20.8% match rate as most collisions occur on roads outside land use boundaries. A 50-metre buffer was therefore applied to unmatched points, improving the match rate to 97.5%. As two-wheeler-specific flow data was unavailable — TfL cycle counts cover only pedal cycles — food establishment density was computed as the count of restaurants, cafes, and fast food outlets within a 500-metre buffer of each collision point, serving as a spatial proxy for gig economy delivery intensity

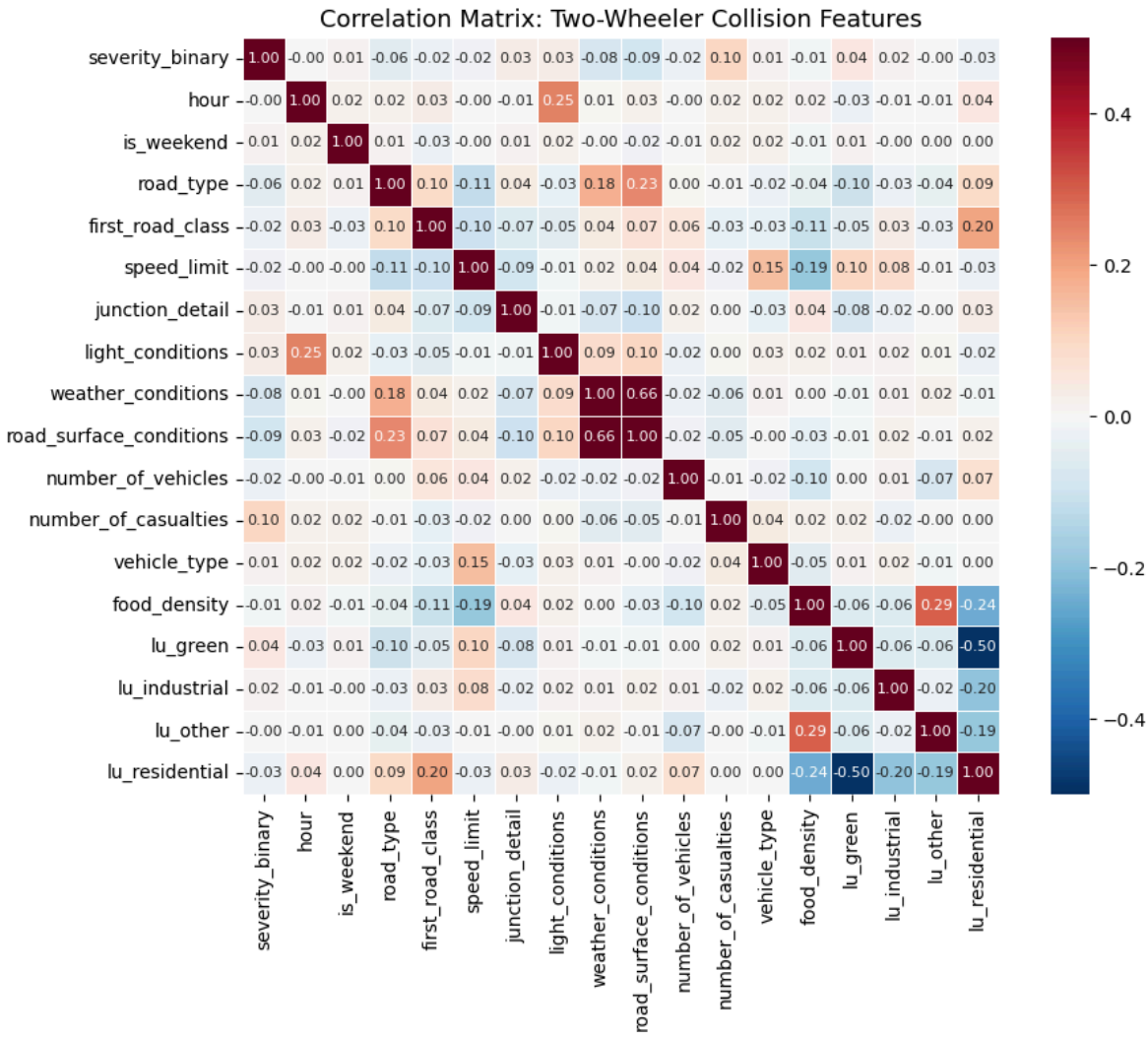
(Christie & Ward, 2023). Multicollinearity was assessed using VIF; urban_or_rural_area was excluded due to a VIF of 35.3, with all retained features below 12.

```
# 4.1 EDA

fig, axes = plt.subplots(2, 3, figsize=(16, 10))
fig.suptitle('Exploratory Analysis: Two-Wheeler Collision Severity in Londo
             fontsize=14, fontweight='bold', y=1.01)

# 4.1.1 Severity rate by hour
hourly = df_model.groupby('hour')['severity_binary'].mean().reset_index()
axes[0,0].bar(hourly['hour'], hourly['severity_binary'], color='steelblue',
axes[0,0].set_title('Serious/Fatal Rate by Hour of Day')
axes[0,0].set_xlabel('Hour'); axes[0,0].set_ylabel('Serious/Fatal Rate')
axes[0,0].yaxis.set_major_formatter(mtick.PercentFormatter(1.0))

# 4.1.2 Severity rate by day of week
day_labels = ['Mon','Tue','Wed','Thu','Fri','Sat','Sun']
daily = df_tw.groupby('day_of_week')['severity_binary'].mean()
axes[0,1].bar(day_labels, daily.values, color='coral', alpha=0.8)
axes[0,1].set_title('Serious/Fatal Rate by Day of Week')
axes[0,1].set_xlabel('Day'); axes[0,1].set_ylabel('Serious/Fatal Rate')
axes[0,1].yaxis.set_major_formatter(mtick.PercentFormatter(1.0))

# 4.1.3 Severity rate by road type
road_labels = {1:'Roundabout', 2:'One way', 3:'Dual carriageway',
               6:'Single carriageway', 7:'Slip road', 9:'Unknown'}
df_model['road_type_label'] = df_model['road_type'].map(road_labels).fillna
road = df_model.groupby('road_type_label')['severity_binary'].mean().sort_v
axes[0,2].barh(road.index, road.values, color='teal', alpha=0.8)
axes[0,2].set_title('Serious/Fatal Rate by Road Type')
axes[0,2].set_xlabel('Serious/Fatal Rate')
axes[0,2].xaxis.set_major_formatter(mtick.PercentFormatter(1.0))

# 4.1.4 Severity rate by speed limit
speed = df_model.groupby('speed_limit')['severity_binary'].mean().reset_ind
axes[1,0].bar(speed['speed_limit'].astype(str), speed['severity_binary'],
              color='mediumpurple', alpha=0.8)
axes[1,0].set_title('Serious/Fatal Rate by Speed Limit')
axes[1,0].set_xlabel('Speed Limit (mph)'); axes[1,0].set_ylabel('Serious/Fa
axes[1,0].yaxis.set_major_formatter(mtick.PercentFormatter(1.0))

# 4.1.5 Severity rate by light conditions
light_labels = {1:'Daylight', 4:'Dark - lit', 5:'Dark - unlit', 6:'Dark - n
df_model['light_label'] = df_model['light_conditions'].map(light_labels).fi
light = df_model.groupby('light_label')['severity_binary'].mean().sort_valu
axes[1,1].barh(light.index, light.values, color='darkorange', alpha=0.8)
axes[1,1].set_title('Serious/Fatal Rate by Light Conditions')
axes[1,1].set_xlabel('Serious/Fatal Rate')
axes[1,1].xaxis.set_major_formatter(mtick.PercentFormatter(1.0))

# 4.1.6 Collision count by hour (volume context)
hour_count = df_model.groupby('hour').size()
axes[1,2].bar(hour_count.index, hour_count.values, color='slategray', alpha
axes[1,2].set_title('Collision Count by Hour of Day')
axes[1,2].set_xlabel('Hour'); axes[1,2].set_ylabel('Number of Collisions')
```

```
plt.tight_layout()
plt.savefig('eda_severity.png', dpi=150, bbox_inches='tight')
plt.show()
```




```
#  4.1.7 Correlation Heatmap

fig, ax = plt.subplots(figsize=(10, 8))

corr_cols = ['severity_binary', 'hour', 'is_weekend',
             'road_type', 'first_road_class', 'speed_limit', 'junction_deta
             'light_conditions', 'weather_conditions',
             'road_surface_conditions',
             'number_of_vehicles', 'number_of_casualties',
             'vehicle_type', 'food_density'] + landuse_cols

corr_matrix = df_model[corr_cols].corr()

sns.heatmap(corr_matrix,
            annot=True, fmt='.2f', cmap='RdBu_r', center=0,
            vmin=-0.5, vmax=0.5,
            square=True, linewidths=0.5, ax=ax,
            annot_kws={'size': 8})

ax.set_title('Correlation Matrix: Two-Wheeler Collision Features', fontsize
plt.tight_layout()
plt.savefig('correlation_heatmap.png', dpi=150, bbox_inches='tight')
plt.show()

print("\nTop correlations with severity_binary:")
print(corr_matrix['severity_binary'].drop('severity_binary').abs()
      .sort_values(ascending=False).head(8).round(3))
```




```
Top correlations with severity_binary:
number_of_casualties       0.104
road_surface_conditions    0.091
weather_conditions         0.080
road_type                  0.055
lu_green                   0.045
junction_detail            0.030
light_conditions           0.029
lu_residential             0.026
Name: severity_binary, dtype: float64
```

Secondly, logistic regression was applied to quantify the direction and strength of each factor's association with severe or fatal consequences. Before model fitting, the SMOTE algorithm was applied to the training set to address class imbalance, generating a balanced training distribution with 3163 samples per class.

```
#   4.2 Logistic Regression

X = df_model[features].copy()

# Ensure all columns are numeric
# The One-Hot column is a boolean value, which should be converted to an in
for col in landuse_cols:
    X[col] = X[col].astype(int)

y = df_model['severity_binary']
```

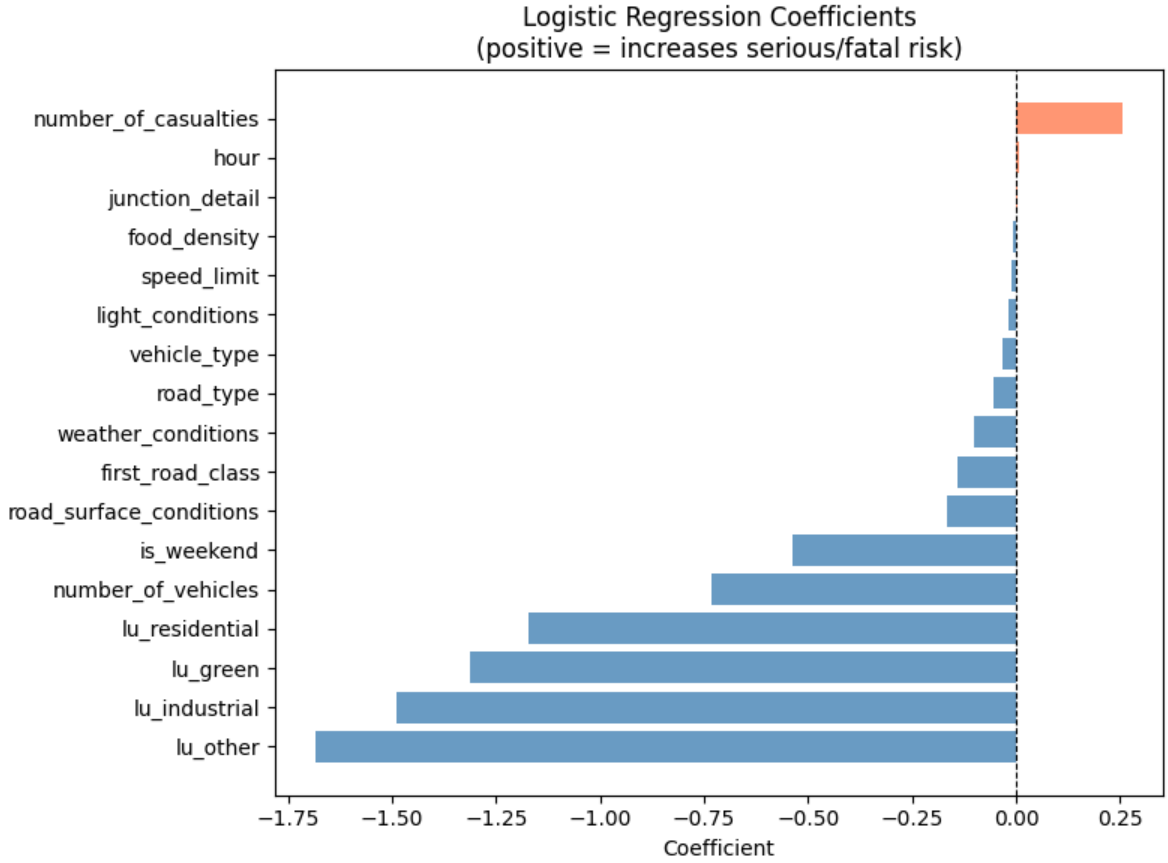
```
print("Feature dtypes:")
print(X.dtypes)
print(f"\nX shape: {X.shape}")

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

#   4.2.1 SMOTE on training set only
smote = SMOTE(random_state=42)
X_train_sm, y_train_sm = smote.fit_resample(X_train, y_train)
print(f"After SMOTE - Training set class distribution: {pd.Series(y_train_s
```

```
Feature dtypes:
hour                        float64
is_weekend                    int64
road_type                     int64
first_road_class              int64
speed_limit                   int64
junction_detail               int64
light_conditions              int64
weather_conditions            int64
road_surface_conditions       int64
number_of_vehicles            int64
number_of_casualties          int64
vehicle_type                  int64
food_density                  int64
lu_green                      int64
lu_industrial                 int64
lu_other                      int64
lu_residential                int64
dtype: object

X shape: (4966, 17)
After SMOTE - Training set class distribution: {0: 3163, 1: 3163}
```

```
#   4.2.2 Train logistic regression
lr = LogisticRegression(max_iter=1000, random_state=42)
lr.fit(X_train_sm, y_train_sm)

#   4.2.3 Evaluate
y_pred_lr = lr.predict(X_test)
y_prob_lr = lr.predict_proba(X_test)[:, 1]

print("\n--- Logistic Regression ---")
print(classification_report(y_test, y_pred_lr, target_names=['Slight','Seri
print(f"ROC-AUC: {roc_auc_score(y_test, y_prob_lr):.3f}")

#   4.2.4 Coefficient plot
coef_df = pd.DataFrame({
    'Feature': features,
    'Coefficient': lr.coef_[0]
}).sort_values('Coefficient')

fig, ax = plt.subplots(figsize=(8, 6))
colors = ['coral' if c > 0 else 'steelblue' for c in coef_df['Coefficient']
ax.barh(coef_df['Feature'], coef_df['Coefficient'], color=colors, alpha=0.8
ax.axvline(0, color='black', linewidth=0.8, linestyle='--')
ax.set_title('Logistic Regression Coefficients\n(positive = increases serio
ax.set_xlabel('Coefficient')
plt.tight_layout()
```

```
plt.savefig('lr_coefficients.png', dpi=150, bbox_inches='tight')
plt.show()
```

```
--- Logistic Regression ---
               precision    recall  f1-score   support

       Slight       0.81      0.70      0.75       791
Serious/Fatal       0.23      0.34      0.28       203

     accuracy                           0.63       994
    macro avg       0.52      0.52      0.51       994
 weighted avg       0.69      0.63      0.65       994

ROC-AUC: 0.527
```

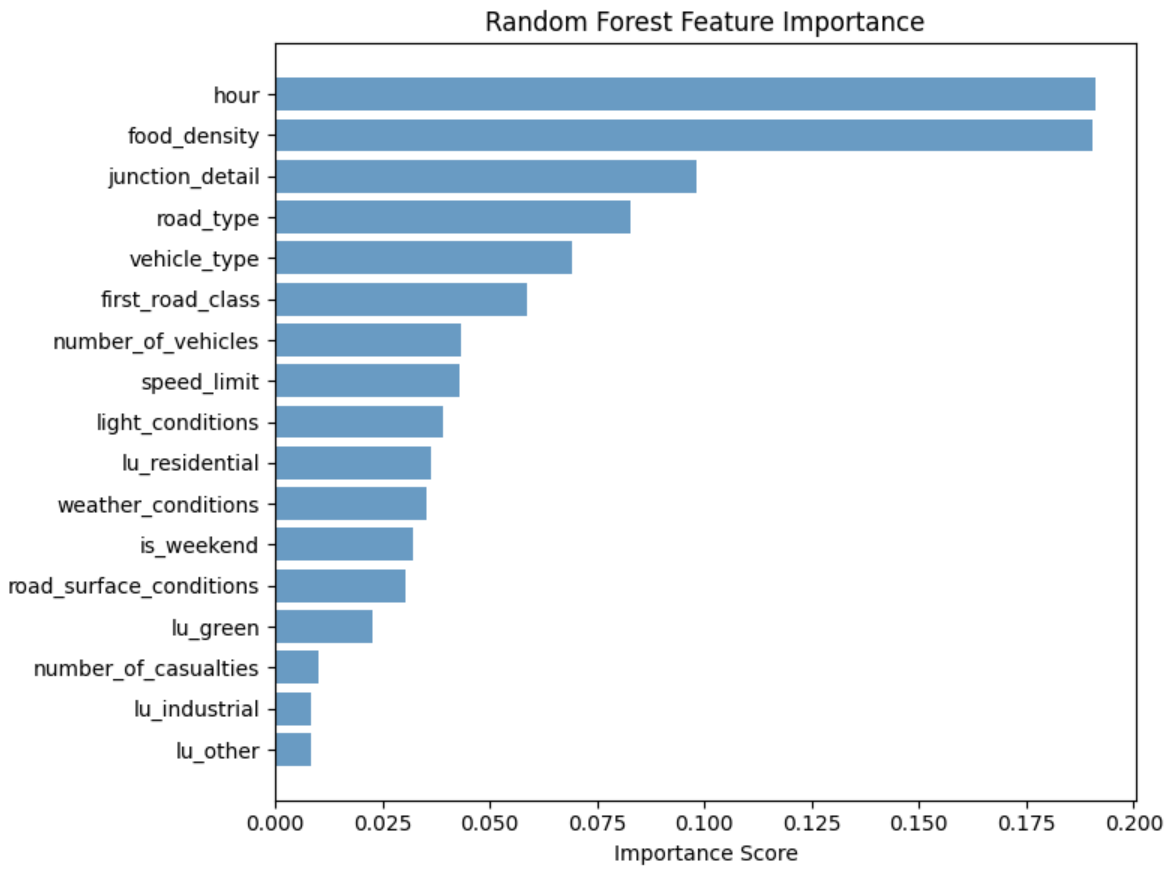


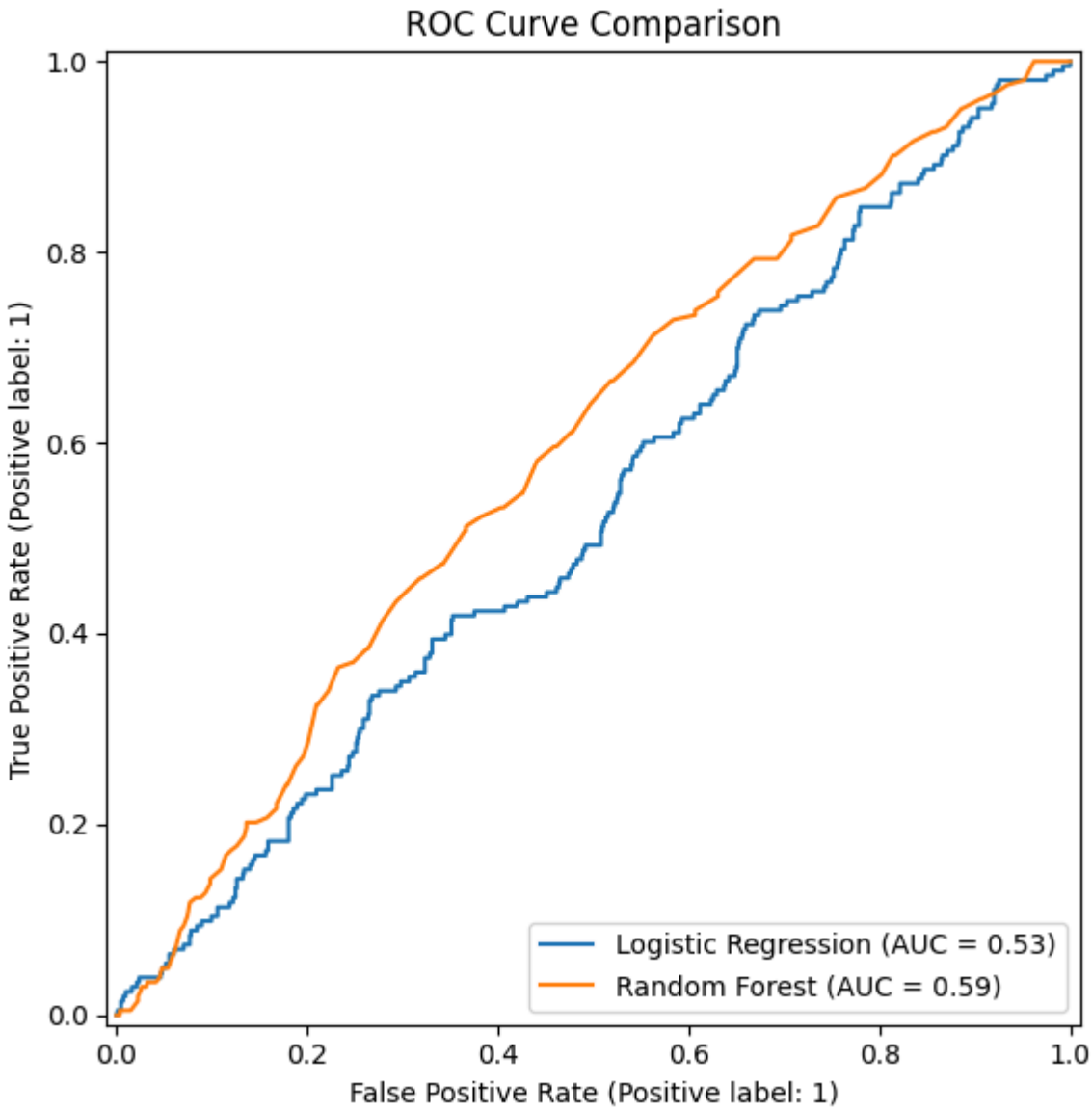
Thirdly, a random forest classifier was trained using the same training/test set split to predict collision severity, and class imbalance was addressed by balancing class weights.The model was trained using 100 estimators without hyperparameter tuning. Feature importance scores were extracted and ranked according to the contribution of the predictor variables to the model performance.

```
#  4.3 Random Forest

rf = RandomForestClassifier(n_estimators=100, random_state=42, class_weight
rf.fit(X_train_sm, y_train_sm)

y_pred_rf = rf.predict(X_test)
y_prob_rf = rf.predict_proba(X_test)[:, 1]

print("--- Random Forest ---")
print(classification_report(y_test, y_pred_rf, target_names=['Slight','Seri
print(f"ROC-AUC: {roc_auc_score(y_test, y_prob_rf):.3f}")
```

```
#   4.3.1Feature importance plot
fi_df = pd.DataFrame({
    'Feature': features,
    'Importance': rf.feature_importances_
}).sort_values('Importance', ascending=True)

fig, ax = plt.subplots(figsize=(8, 6))
ax.barh(fi_df['Feature'], fi_df['Importance'], color='steelblue', alpha=0.8
ax.set_title('Random Forest Feature Importance')
ax.set_xlabel('Importance Score')
plt.tight_layout()
plt.savefig('rf_importance.png', dpi=150, bbox_inches='tight')
plt.show()

#   4.3.2 ROC curve comparison
fig, ax = plt.subplots(figsize=(7, 6))
RocCurveDisplay.from_predictions(y_test, y_prob_lr, name='Logistic Regressi
RocCurveDisplay.from_predictions(y_test, y_prob_rf, name='Random Forest', a
ax.set_title('ROC Curve Comparison')
plt.tight_layout()
plt.savefig('roc_comparison.png', dpi=150, bbox_inches='tight')
plt.show()
```

```
--- Random Forest ---
               precision    recall  f1-score   support

       Slight       0.81      0.86      0.83       791
Serious/Fatal       0.28      0.20      0.23       203

     accuracy                           0.73       994
    macro avg       0.54      0.53      0.53       994
 weighted avg       0.70      0.73      0.71       994

ROC-AUC: 0.593
```


Random Forest Feature Importance
hour
food_density
junction_detail
road_type
vehicle_type
first_road_class
number_of_vehicles
speed_limit
light_conditions
lu_residential
weather_conditions
is_weekend
road_surface_conditions
lu_green
number_of_casualties
lu_industrial
lu_other
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Importance Score

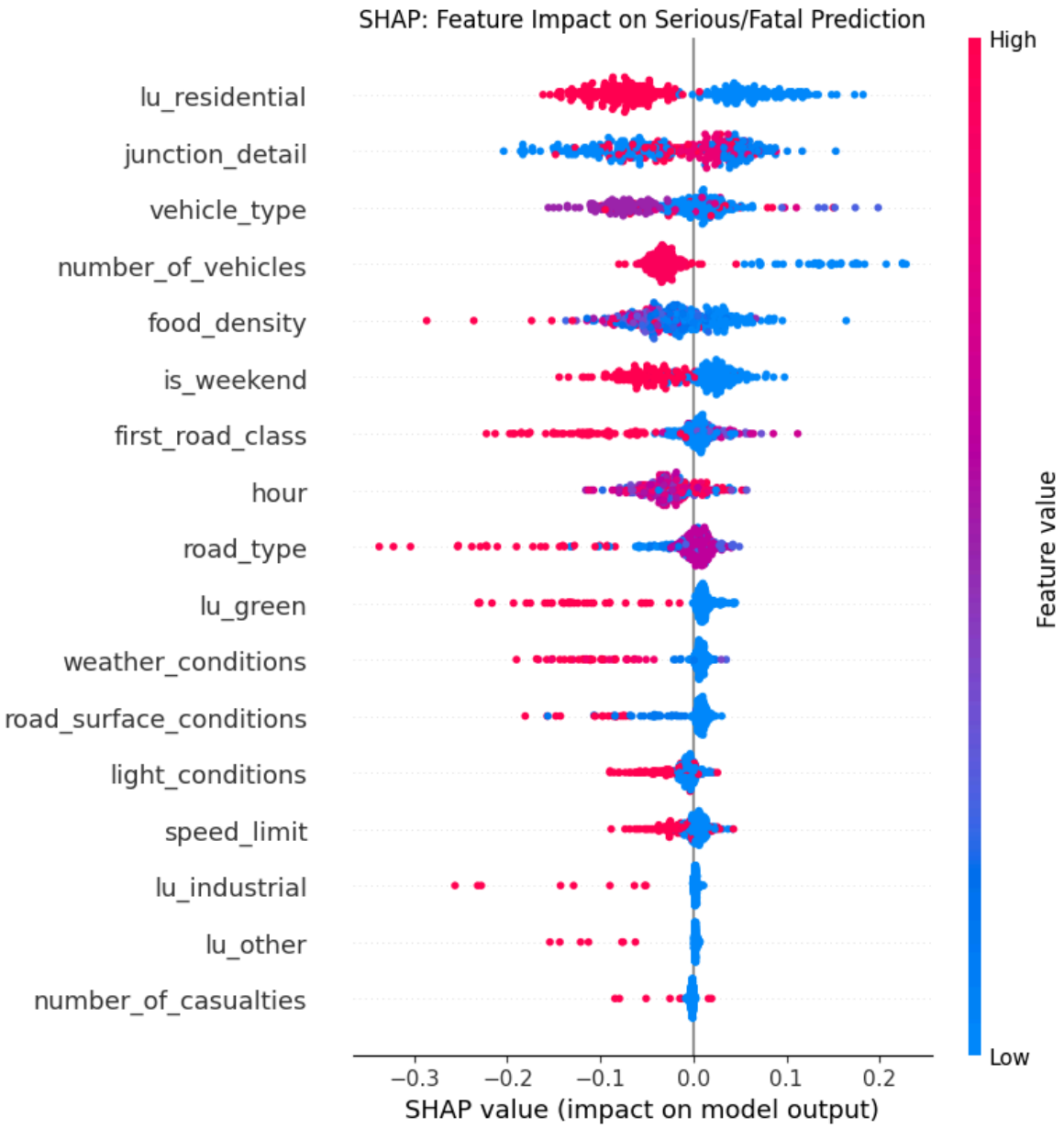


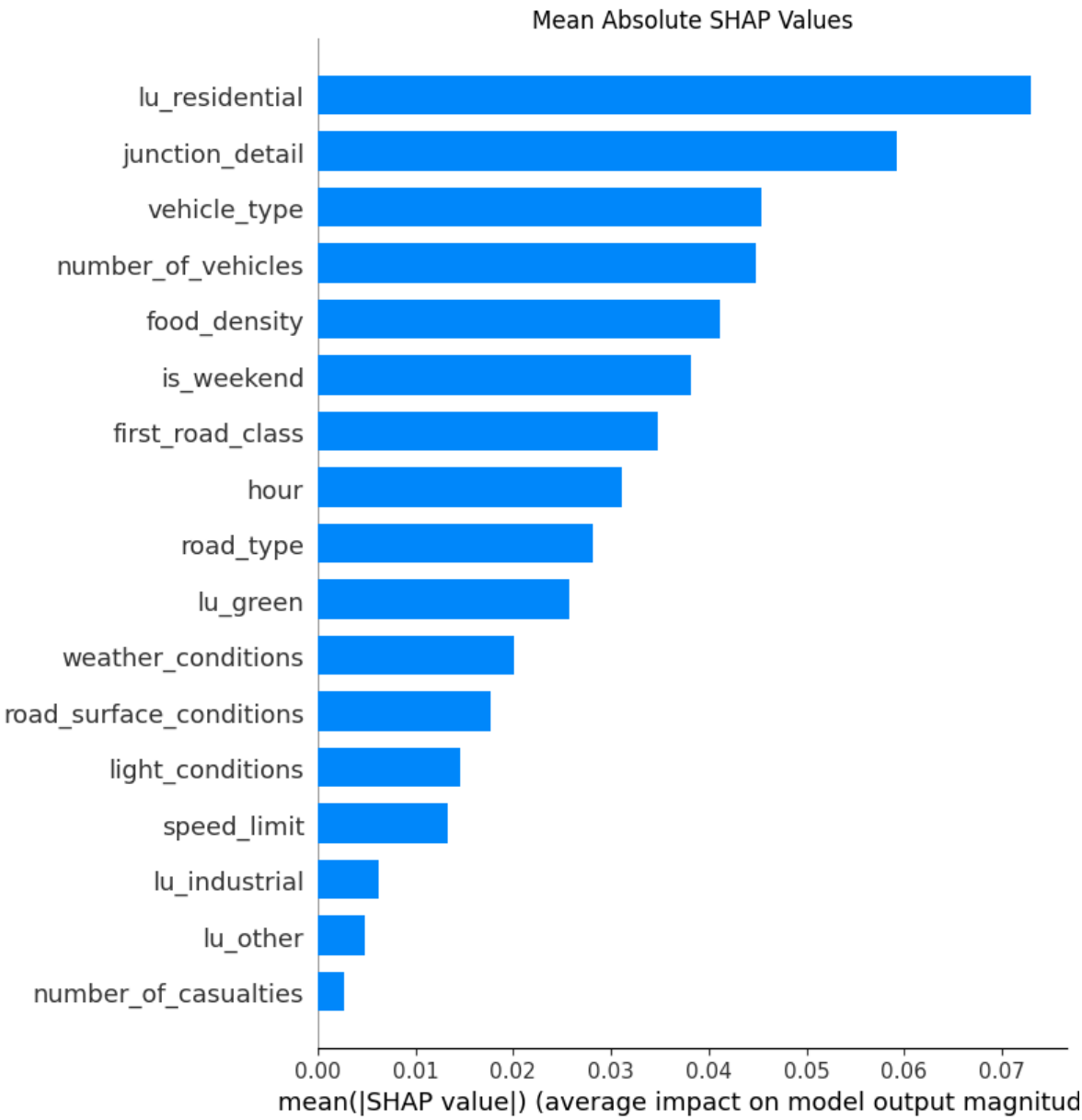
Fourth, we calculated the SHAP value to interpret the random forest model, revealing which features are most important and the direction and strength of their influence on the prediction of severe or fatal consequences. To improve computational efficiency, we used a random sample containing 300 test observations, with a fixed random seed of 42 to ensure the reproducibility of the results.

All models were evaluated using ROC-AUC, precision, recall, and F1 score on a reserved test set containing 20% of the data.

```
#  4.4 SHAP Analysis

explainer = shap.TreeExplainer(rf)
X_test_sample = X_test.sample(300, random_state=42)
shap_values = explainer.shap_values(X_test_sample)

# Adapted to the output format of SHAP 0.51.0
if isinstance(shap_values, list):
    shap_vals = shap_values[1]  # class 1 = Serious/Fatal
else:
```

```
    shap_vals = shap_values[:, :, 1]

#  4.4.1 Beeswarm plot
plt.figure()
shap.summary_plot(shap_vals, X_test_sample,
                  feature_names=features, show=False)
plt.title('SHAP: Feature Impact on Serious/Fatal Prediction')
plt.tight_layout()
plt.savefig('shap_summary.png', dpi=150, bbox_inches='tight')
plt.show()

#  4.4.2 Bar plot
plt.figure()
shap.summary_plot(shap_vals, X_test_sample,
                  feature_names=features, plot_type='bar', show=False)
plt.title('Mean Absolute SHAP Values')
plt.tight_layout()
plt.savefig('shap_bar.png', dpi=150, bbox_inches='tight')
plt.show()
```





5,0 Results and Discussion

5.1 Exploratory Data Analysis

EDA revealed two distinct temporal risk profiles: the highest serious rate occurs between 22:00 and 05:00, exceeding 30% at certain hours, while the greatest absolute volume of serious collisions falls in weekday evenings (17:00–22:00), coinciding with peak gig economy delivery activity (Christie & Ward, 2023). This distinction is important for intervention design — deep-night collisions are most likely fatal, but weekday evenings represent the greatest overall burden. Dual carriageways show the highest serious rate , and 70mph roads approach 38%, consistent with speed-severity literature (Antonella et al., 2023). The correlation heatmap shows weak negative correlations for road surface conditions ($r = -0.09$) and weather conditions ($r = -0.08$), reflecting STATS19 encoding convention: higher numeric codes represent more adverse conditions, so the negative direction indicates that dry conditions are associated with higher severity, likely because riders travel faster in good weather.

5.2 Logistic Regression

The logistic regression model achieved ROC-AUC of 0.527. number_of_casualties ($\beta = +0.21$) and light_conditions ($\beta \approx +0.05$) show positive associations with serious outcomes. All land use dummies show negative coefficients relative to the commercial baseline — lu_other ($\beta = -2.1$) and lu_industrial ($\beta = -1.5$) are largest — suggesting commercial road environments carry comparatively higher severity risk. The negative is_weekend coefficient indicates weekday collisions are more severe, consistent with concentrated gig economy delivery activity on weekday evenings (Christie & Ward, 2023).

5.3 Random Forest and Feature Importance

The random forest achieved ROC-AUC of 0.593, outperforming logistic regression and better capturing non-linear relationships. The improvement over the baseline model (AUC = 0.562, without vehicle_type, first_road_class, and food_density) reflects the predictive value of these additional variables. lu_residential dominates feature importance, followed by junction_detail, vehicle_type, number_of_vehicles, and food_density. urban_or_rural_area was excluded due to extreme multicollinearity (VIF = 35.3) and negligible predictive contribution.

5.4 SHAP Analysis

SHAP analysis provides the most granular answer to RQ2. lu_residential records the highest mean absolute SHAP value, followed by junction_detail, vehicle_type, number_of_vehicles, and food_density. hour ranks eighth, indicating that once vehicle type, road hierarchy, and spatial exposure are accounted for, the marginal contribution of time of day is reduced.

High food_density values are associated with negative SHAP values, indicating lower predicted severity in commercially dense areas — typically zones with lower operating speeds and stricter traffic management. This supports a key distinction: high-density commercial areas generate more two-wheeler collisions due to greater delivery exposure, but incidents tend to be less severe. The weekday evening collision peak (17:00–22:00), combined with elevated motorcycle proportions in high food_density zones, indirectly supports the gig rider exposure hypothesis (Christie & Ward, 2023).

lu_residential collisions show negative SHAP values relative to commercial zones, yet lu_residential ranks as the most important predictor overall. This reflects the risk profile of arterial roads bordering residential areas, where higher speeds and less infrastructure protection elevate severity when collisions do occur. vehicle_type shows mixed SHAP distributions, confirming that motorcyclists face substantially different severity outcomes compared to cyclists (DfT, 2025).

Both models achieve modest ROC-AUC values (logistic regression: 0.527, random forest: 0.593), reflecting unobserved factors such as rider behaviour, helmet use, and vehicle condition absent from police-recorded data (Antonella et al., 2023). The highest-risk scenario involves motorcycle incidents at junctions in residential or mixed-use areas during weekday evenings — precisely the operational context of gig economy delivery riders (Christie & Ward, 2023).

6.0 Conclusion

This study used 4,966 collision records from the UK Department for Transport 2025 provisional dataset, combined with OSM land use data and catering establishment density information, to analyze temporal, road infrastructure, land use, and vehicle-level factors related to the severity of two-wheeler collisions in London.

For RQ1, the study found that intersection type, land use, vehicle type, and catering establishment density were key factors. Two distinct temporal risk characteristics were identified: collision severity was highest late at night (22:00–05:00), while the absolute value of severe injuries was highest on weekday evenings (17:00–22:00), coinciding with peak delivery times in the gig economy. For RQ2, SHAP analysis confirmed that lu_residential was the most influential single predictor, followed by junction_detail and vehicle_type. High food density was associated with lower accident severity, supporting the distinction between collision frequency and severity in densely populated commercial areas.

These findings suggest that redesigning intersections, implementing speed controls on residential arterial roads, and enacting safety programs for gig economy delivery platforms on weekday evenings hold the greatest potential for reducing serious and fatal collisions.

Both models exhibited low ROC-AUC values (logistic regression: 0.527, random forest: 0.593), reflecting unobserved factors not included in police records, such as cyclist behavior and helmet usage. OSM spatial connectivity relies on a 50-meter buffer approximation, with food density serving only as a proxy for two-wheeled vehicle traffic rather than a direct measurement. Future research could incorporate real-time bicycle traffic data and bicycle infrastructure density as additional spatial predictors.

7.0 References

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